THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 6	Chord $= 2\sqrt{r^2 - d^2}$.
2	**(b)** 22	Arc $= \frac{\theta}{360} \times 2\pi r$.
3	**(b)** $\frac{1}{6}$	There are 36 equally likely outcomes.
4	**(a)** 4	$d = \frac{t_n - t_1}{n-1} = \frac{40}{10}$.
5	**(b)** 5	$r^2 = h^2 + d^2$.
6	**(a)** 12	$h = \frac{2A}{b}$.
7	**(d)** $\frac{1}{6}$	Count favourable outcomes out of 36.
8	**(a)** 9	$a = t_n - (n-1)d = 69 - 60$.
9	**(d)** 58°	Isosceles triangle OAB.
10	**(d)** 12	$h = \frac{A}{b}$.
11	**(d)** $\frac{2}{3}$	Count the outcomes in either event, without double counting.
12	**(a)** 59	$5d = 20$ so $d = 4$; then $t_{15} = t_9 + 6d$.
13	**(c)** 72°	$\angle A + \angle C = 180^\circ$.
14	**(d)** 52	Side $= \sqrt{A}$.
15	**(a)** $\frac{7}{12}$	Both favourable and total counts increase.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Half-chord $= \sqrt{20^2 - 16^2} = 12$ cm (1); Chord $= 24$ cm (1).	1+1
17	$s = 32$, $A = 168$ cm^2 (1); $h = \frac{2 \times 168}{14} = 24$ cm (1).	1+1
18	Outcomes with sum 6: (1,5), (2,4), (3,3), (4,2), (5,1) (1); $P = \frac{5}{36} = \frac{5}{36}$ (1).	1+1
19	$(q - p)d = 30$ (1); $d = 6$ (1).	1+1
20	$\angle C = 94^\circ$ (1); $\angle D = 81^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Distances from the centre: 15 cm and 8 cm (1); Same side: subtract (1); Distance $= 7$ cm (1).	1+1+1
22	$s = 20$ cm (1); $s - a = 16$, $s - b = 12$, $s - c = 9$, $s - d = 3$ (1); $A = 72$ cm^2 (1).	1+1+1
23	$P(\text{a number less than } 3) = \frac{1}{3}$ (1); $P(\text{a number greater than } 4) = \frac{1}{3}$ (1); Favourable for either: 4, so $P = \frac{2}{3}$ (1).	1+1+1
24	$n = \frac{l - a}{d} + 1 = 25$ (1); $S = \frac{n}{2}(a + l)$ (1); $S = 2000$ (1).	1+1+1
	OR: $\angle A + \angle C = 180^\circ$; $5x = 170$ (1); $x = 34$ (1); $\angle A = 122^\circ$, $\angle C = 58^\circ$ (1).	1+1+1
25	Perimeter $= 2 \times 30 + 14 + \frac{1}{2}\pi \times 14 = 74 + 22$; $= 96$ cm (1); Area $= 420 + \frac{1}{2}\pi \left(\frac{14}{2}\right)^2$ (1); $= 497$ cm^2 (1).	1+1+1
	OR: Likes tea $= 70 + 30 = 100$; $P(\text{tea}) = \frac{100}{200} = \frac{1}{2}$ (1); Neither $= 200 - 70 - 50 - 30 = 50$ (1); $P(\text{neither}) = \frac{1}{4}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Half-chord $= 12$ cm (1); Distance $d = \sqrt{13^2 - 12^2} = 5$ cm (1); Area $= \frac{1}{2} \times 24 \times 5$ (1); $= 60$ cm^2 (1); Base is the chord, height is the perpendicular (1).	1+1+1+1+1
	OR: Plot area $= 1200$ m^2 (1); Pond area $= \frac{22}{7} \times 7^2 = 154$ m^2 (1); Remaining $= 1046$ m^2 (1); Cost $= 1046 \times 6$ (1); $=$ ₹6276 (1).	1+1+1+1+1
27	Sum 5: 4 outcomes; sum 9: 4 outcomes (1); $P(5) = \frac{1}{9}$ (1); $P(9) = \frac{1}{9}$ (1); $P(\text{either}) = \frac{8}{36} = \frac{2}{9}$ (2).	1+1+1+2
	OR: Lengths form an AP with $d = -4$ (1); Top rung $= 120 - 10 \times 4 = 80$ cm (1); $S = \frac{n}{2}(a + l)$ (1); $S = \frac{11}{2}(120 + 80)$ (1); Total 1100 cm (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 182	$2+4+\cdots+2n=n(n+1)$.
2	**(b)** 48°	Central angle is double.
3	**(b)** 140	Area $=\frac{1}{2}d(h_1+h_2)$.
4	**(c)** $\frac{4}{11}$	Count the matching letters, including repeats.
5	**(b)** 196	$1+3+5+\cdots+(2n-1)=n^2$.
6	**(b)** 135°	Opposite angles of the cyclic quadrilateral are supplementary.
7	**(c)** 126	Find the breadth first.
8	**(b)** 7200	Expected number = total × probability.
9	**(a)** 3	$t_5/t_3=r^2$.
10	**(d)** 12	Equal chords are equidistant from the centre.
11	**(a)** 11	Perimeter is unchanged.
12	**(a)** $\frac{7}{12}$	Use the complement rule.
13	**(b)** 19	These form an AP; count with $n=\frac{l-a}{d}+1$.
14	**(c)** 66°	Right angle at C.
15	**(a)** 110	Distance $=n\times$ circumference.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$A=\frac{1}{2}\times26\times9+\frac{1}{2}\times26\times7$ (1); $A=208$ cm^2 (1).	1+1
17	New red $=9$, new total $=18$ (1); $P=\frac{1}{2}$ (1).	1+1
18	First 15, last 100; $n=18$ (1); Sum $=1035$ (1).	1+1
19	$\angle OAB=\frac{180-68}{2}=56^\circ$ (1); $\angle ACB=34^\circ$ (1).	1+1
20	Arc $=\frac{40}{360}\times2\times\frac{22}{7}\times63$ (1); $=44$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$d=5$ (1); $a=6$ (1); $t_{20}=101$ (1).	1+1+1
22	$\angle OAB=\frac{180^\circ-72^\circ}{2}=54^\circ$ (1); $\angle ACB=\frac{1}{2}\angle AOB=36^\circ$ (1); $\angle ADB=180^\circ-36^\circ=144^\circ$ (1).	1+1+1
23	$A=15\times8=120$ cm^2 (1); $h'=\frac{120}{12}=10$ cm (2).	1+2
24	Let x red balls be added: $\frac{3+x}{12+x}=\frac{1}{2}$ (1); $2(3+x)=1(12+x)$ (1); $x=6$ (1).	1+1+1
	OR: $d=\frac{26-5}{7}=3$ (1); $S_n=\frac{n}{2}[2a+(n-1)d]$ (1); $S_{20}=670$ (1).	1+1+1
25	Distances from the centre: 24 cm and 7 cm (1); Same side: subtract (1); Distance $=17$ cm (1).	1+1+1
	OR: $s=24$ cm (1); $s-a=16$, $s-b=15$, $s-c=12$, $s-d=5$ (1); $A=120$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$d_1=\sqrt{13^2-5^2}=12$ cm (1); $d_2=\sqrt{13^2-12^2}=5$ cm (1); The chord of length 24 cm is nearer to the centre (1); A longer chord lies nearer to the centre (2).	1+1+1+2
	OR: $s=18$ m (1); $A=\sqrt{s(s-a)(s-b)(s-c)}=60$ m^2 (1); Cost $=60\times5$ (1); $=$ ₹300 (1); Units stated (1).	1+1+1+1+1
27	$P(\text{defective})=\frac{8}{200}=\frac{1}{25}$ (1); Expected defective $=5000\times\frac{1}{25}$ (1); $=200$ (1); Units stated (2).	1+1+1+2
	OR: Each year multiplies by $110/100$ (1); Hence a GP with this common ratio (1); Amount after n years $=P(1+\frac{r}{100})^n$ (1); $=10000\times(1.1)^3$ (1); Amount ₹13310 (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{1}{2}$	Use the 8 outcomes.
2	**(d)** 70	$t_{20} - t_{10} = 10d$.
3	**(c)** 12	Chord $= 2\sqrt{r^2 - d^2}$.
4	**(b)** 88	Arc $= \frac{\theta}{360} \times 2\pi r$.
5	**(d)** $\frac{1}{9}$	There are 36 equally likely outcomes.
6	**(c)** -3	$d = \frac{t_n - t_1}{n-1} = \frac{-24}{8}$.
7	**(b)** 10	$r^2 = h^2 + d^2$.
8	**(d)** 8	$h = \frac{2A}{b}$.
9	**(a)** $\frac{1}{6}$	Count favourable outcomes out of 36.
10	**(a)** 30	$a = t_n - (n-1)d = 16 - -14$.
11	**(d)** 48°	Isosceles triangle OAB.
12	**(b)** 8	$h = \frac{A}{b}$.
13	**(a)** $\frac{2}{3}$	Count the outcomes in either event, without double counting.
14	**(b)** -8	$5d = -10$ so $d = -2$; then $t_{15} = t_9 + 6d$.
15	**(a)** 80°	$\angle A + \angle C = 180^\circ$.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$P(\text{not win}) = 1 - \frac{2}{5} = \frac{3}{5}$ (1); Expected wins $= 200 \times \frac{2}{5} = 80$ (1).	1+1
17	$r^3 = 27$ (1); $r = 3$, $a = 2$ (1).	1+1
18	Half-chord $= \sqrt{25^2 - 15^2} = 20$ cm (1); Chord $= 40$ cm (1).	1+1
19	$s = 27$, $A = 126$ cm^2 (1); $h = \frac{2\times126}{21} = 12$ cm (1).	1+1
20	Outcomes with sum 8: (2,6), (3,5), (4,4), (5,3), (6,2) (1); $P = \frac{5}{36} = \frac{5}{36}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$P(\text{a prime number}) = \frac{1}{2}$ (1); $P(\text{a multiple of 3}) = \frac{1}{3}$ (1); Favourable for either: 4, so $P = \frac{2}{3}$ (1).	1+1+1
22	$n = \frac{l-a}{d} + 1 = 20$ (1); $S = \frac{n}{2}(a + l)$ (1); $S = 690$ (1).	1+1+1
23	$\angle A + \angle C = 180^\circ$; $6x = 174$ (1); $x = 29$ (1); $\angle A = 128^\circ$, $\angle C = 52^\circ$ (1).	1+1+1
24	Perimeter $= 2 \times 25 + 28 + \frac{1}{2}\pi \times 28 = 78 + 44$; $= 122$ cm (1); Area $= 700 + \frac{1}{2}\pi \left(\frac{28}{2}\right)^2$ (1); $= 1008$ cm^2 (1).	1+1+1
	OR: Likes tea $= 110 + 40 = 150$; $P(\text{tea}) = \frac{150}{300} = \frac{1}{2}$ (1); Neither $= 300 - 110 - 60 - 40 = 90$ (1); $P(\text{neither}) = \frac{3}{10}$ (1).	1+1+1
25	$d = -4$ (1); $a = 30$ (1); $t_{14} = -22$ (1).	1+1+1
	OR: $\angle OAB = \frac{180^\circ - 88^\circ}{2} = 46^\circ$ (1); $\angle ACB = \frac{1}{2}\angle AOB = 44^\circ$ (1); $\angle ADB = 180^\circ - 44^\circ = 136^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Half-chord $= 12$ cm (1); Distance $d = \sqrt{15^2 - 12^2} = 9$ cm (1); Area $= \frac{1}{2} \times 24 \times 9$ (1); $= 108$ cm^2 (1); Base is the chord, height is the perpendicular (1).	1+1+1+1+1
	OR: Plot area $= 2000$ m^2 (1); Pond area $= \frac{22}{7} \times 14^2 = 616$ m^2 (1); Remaining $= 1384$ m^2 (1); Cost $= 1384 \times 4$ (1); $=$ ₹5536 (1).	1+1+1+1+1
27	Sum 4: 3 outcomes; sum 10: 3 outcomes (1); $P(4) = \frac{1}{12}$ (1); $P(10) = \frac{1}{12}$ (1); $P(\text{either}) = \frac{6}{36} = \frac{1}{6}$ (2).	1+1+1+2
	OR: Lengths form an AP with $d = -3$ (1); Top rung $= 90 - 14 \times 3 = 48$ cm (1); $S = \frac{n}{2}(a + l)$ (1); $S = \frac{15}{2}(90 + 48)$ (1); Total 1035 cm (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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